Closed-Loop DC Motor Speed Control Using ESP32

This report presents the design and implementation of a closed-loop motor speed control system using an ESP32 microcontroller. The system regulates the speed of a DC geared motor through feedback obtained from an incremental encoder. Multiple user input methods are implemented, including a keypad, a potentiometer, and Bluetooth communication, making the system flexible and suitable for educational and practical mechatronics applications.

1. Introduction

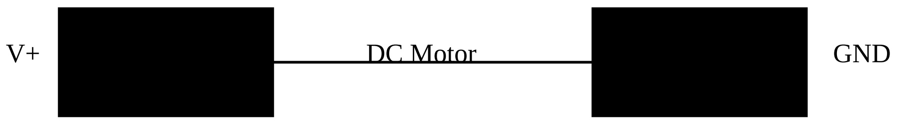
Speed control of DC motors is a critical requirement in many engineering systems such as robotics, industrial automation, electric vehicles, and conveyor mechanisms. Open-loop systems suffer from speed variations due to load changes and supply fluctuations. To overcome these limitations, closed-loop control systems are used. In this project, the ESP32 continuously measures motor speed and automatically adjusts the control signal to maintain the desired speed.

2. Motor Description

The motor used in this project is a DC geared motor (JG25-370-CE type). This motor is well suited for speed control applications due to its compact size, high torque output, and stable operation. The integrated gearbox reduces speed while increasing torque, allowing precise control at low and medium speeds. The motor shaft is coupled with an incremental encoder, enabling accurate speed feedback for closed-loop operation.

3. H-Bridge Motor Driver

An H-Bridge motor driver such as the L298N or L238 is used to interface the low-power ESP32 with the higher-current DC motor. The H-Bridge allows the motor to be driven safely by controlling the direction and speed of rotation. In this project, two digital pins control the motor direction, while a PWM signal from the ESP32 controls the motor speed through the enable pin.



4. Potentiometer Control Mode

In addition to digital speed control, the system provides an analog control mode using a potentiometer connected to the ESP32 ADC pin. The potentiometer allows the user to manually adjust the desired motor speed by rotating its knob. The ESP32 reads the analog voltage value and maps it to a corresponding RPM range. This feature is particularly useful for testing, calibration, and real-time manual control.

When the potentiometer mode is selected, the ESP32 continuously updates the target speed according to the potentiometer position. Despite being a manual input, the system remains closed-loop because the actual motor speed is still measured by the encoder and corrected automatically using feedback control.

5. Closed-Loop Control Principle

This project is a closed-loop control system because it uses feedback from the motor to regulate its speed. The encoder generates pulses proportional to the motor rotation. The ESP32 counts these pulses and calculates the actual motor speed in RPM. The calculated speed is compared with the target speed set by the user. The difference between these values, known as the error, is used to adjust the PWM duty cycle applied to the motor.

A proportional control algorithm is implemented in software. If the motor runs slower than the desired speed, the ESP32 increases the PWM duty cycle. If the motor runs faster, the PWM duty cycle is reduced. This continuous correction ensures stable and accurate speed control under varying load conditions.

6. Software and Code Explanation

The program is written using the Arduino framework for ESP32. Hardware initialization includes setting up GPIO pins, PWM channels, LCD communication, Bluetooth serial communication, and encoder interrupts. The main loop handles user inputs, calculates motor speed, applies the closed-loop control algorithm, and updates the LCD display.

7. Pin Connections

Component	ESP32 Pin	Function
Motor Driver ENA	GPIO 23	PWM speed control
Motor Driver IN1	GPIO 18	Motor direction
Motor Driver IN2	GPIO 19	Motor direction
Encoder Channel A	GPIO 5	Speed feedback (interrupt)
Encoder Channel B	GPIO 13	Direction detection
Potentiometer	GPIO 34	Analog speed reference
LCD SDA	GPIO 21	I2C data
LCD SCL	GPIO 22	I2C clock

8. Conclusion

The ESP32-based motor speed control system demonstrates an effective application of closed-loop control in mechatronics engineering. The integration of encoder feedback ensures accurate speed regulation, while the inclusion of keypad, potentiometer, and Bluetooth control enhances system flexibility. This project successfully combines embedded systems, control theory, and power electronics into a practical and educational engineering solution.